

JAVA Programming Learning Guide

This guide is designed for beginners who want to learn Java programming step-by-step and become ready for Java Full Stack Development and coding interviews.

1. Introduction to Java

- What is Java?
- Features of Java
- JDK, JRE, and JVM
- Installing Java and IDE setup

2. Java Basics

- Variables and Data Types
- Operators
- Input and Output
- Conditional Statements
- Loops

3. Object-Oriented Programming (OOP)

- Classes and Objects
- Constructors
- Inheritance
- Polymorphism
- Abstraction
- Encapsulation

4. Advanced Java Concepts

- Arrays and Strings
- Collections Framework
- Exception Handling
- File Handling
- Multithreading

5. Java for Backend Development

- JDBC
- Servlets
- JSP
- Spring Framework
- Spring Boot
- REST APIs

6. Database

- MySQL Basics
- Connecting Java with MySQL
- CRUD Operations

7. DSA with Java

- Arrays
- Linked Lists
- Stacks and Queues
- Trees and Graphs
- Sorting and Searching

8. Projects

- Student Management System
- Library Management System
- E-Learning Website
- Fix It Now Service Platform

Recommended Learning Timeline

Month	Focus Area
1	Java Basics + OOP
2	Collections + Exception Handling + File Handling
3	DSA with Java
4	JDBC + MySQL + Mini Projects
5	Servlets + JSP + Spring
6	Spring Boot + REST APIs + Full Stack Project

Recommended Tools & Platforms:

- IntelliJ IDEA / VS Code / Eclipse
- MySQL Workbench
- Git & GitHub
- Maven
- Postman

Practice coding daily, build projects, and solve DSA problems consistently. Java combined with Spring Boot and DSA can help you prepare for software engineering jobs and full stack development roles.